

7 Network Epidemics

Diseases spread through contact between people so, studying spreading processes on social networks can be quite informative. These models apply not just to disease modelling, but also to computer viruses on the web or ‘misinformation’ on social media.

Reading

[1] **Chapter 10**, [2] **Chapter 17**.

7.1 SI Model

We start with some simplifying assumptions and study models of infection spreading in a homogeneous population before introducing networks.

- An individual can be in one of the following states at any time
 - Susceptible (S) - able to be infected
 - Infected (I) - is infected
 - Recovered (R) - recovered from the infection and not-infectious.
- The population is homogeneous, so anyone can infect anyone else.

Some diseases you can only get once (chickenpox) so R individuals become invisible to the disease. Some diseases you can get multiple times (flu) so R individuals can transition back the S state. ‘Recovered’ as a description also might be optimistic, dead individuals function the same as recovered ones as far as the model is concerned!

Transition rates and allowed states are based on the biology of disease (or the psychology of misinformation, effectiveness of the phishing email etc.). The homogeneity assumption is the one that will be of concern to us as network scientists. We will can think of homogeneity as equivalent to a contact network which is a complete graph. Eventually we will replace this complete graph with a more realistic social network but for now we work with this assumption to build some intuition about the model behaviours.

Let’s say that individuals have β randomly chosen contacts in a given unit of time with anyone in the population of interest. In a population of N people with S susceptible and I infected ($N = S + I$) a random person is susceptible with probability S/N . Therefore, an infected person contacts

$\beta SI/N$ uninfected people per unit time. Since there are I infected individuals there are $\beta SI/N$ infected-susceptible contacts, each of which changes the state from S to I . We can express this as

$$\frac{dI}{dt} = \beta \frac{SI}{N}, \quad \frac{dS}{dt} = -\beta \frac{SI}{N} \quad (1)$$

It is convenient to use the lower case letter to denote the population normalised quantity i.e. $s = S/N, i = I/N, r = R/N$ then

$$\frac{ds}{dt} = -\beta si, \quad \frac{di}{dt} = \beta si \quad (2)$$

and, since $s + i = 1$,

$$\frac{di}{dt} = \beta(1 - i)i \quad (3)$$

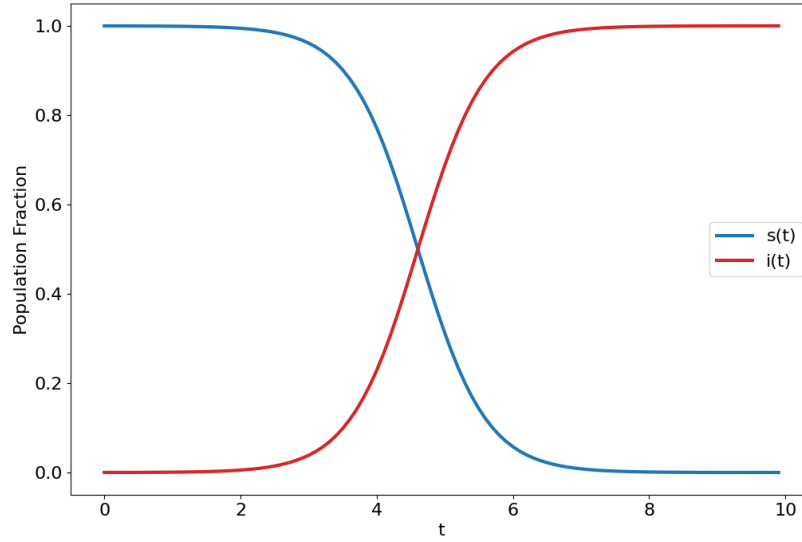


Figure 1: Solution of the SI model with $\beta = 2$.

This is a well known equation called the logistic growth equation, which is solved by the function

$$i(t) = \frac{i_0 e^{\beta t}}{1 - i_0 + i_0 e^{\beta t}} \quad (4)$$

called a **sigmoid** and shown in Figure 1. Actually the solution in Figure 1 was obtained by numerically integrating the model equations which is what we will generally do. Only in the simplest cases can we obtain an analytic solution.

7.2 SIS model

In this model individuals recover at a fixed rate γ and become susceptible again, so we have

$$\frac{ds}{dt} = -\beta si + \gamma i, \quad \frac{di}{dt} = \beta si - \gamma i \quad (5)$$

The same trick $s + i = 1$ allows us to solve this analytically, producing another logistic curve

$$i(t) = \left(1 - \frac{\gamma}{\beta}\right) \frac{Ce^{(\beta-\gamma)t}}{1 + Ce^{(\beta-\gamma)t}} \quad (6)$$

where C is set by the initial condition $i(0) = i_0$.

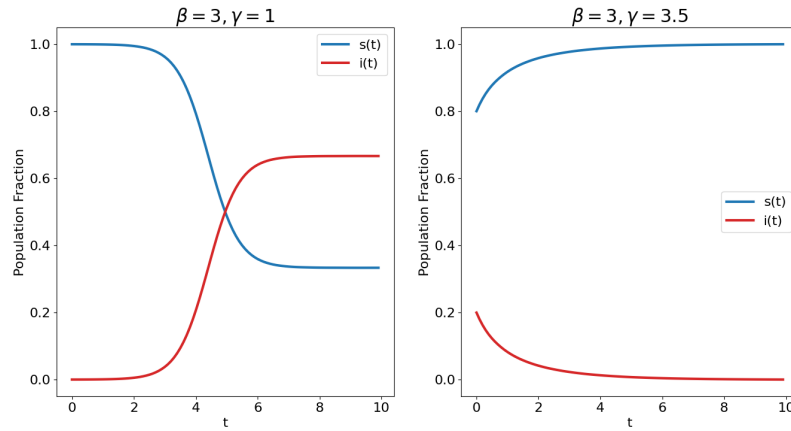


Figure 2: Solution of the SI model with $\beta = 3$, $\gamma < \beta$ and $\gamma > \beta$.

Again we actually just integrate it numerically and Figure 2 shows an example. The most important difference between SI and SIS models is that in the SIS model the whole population does not get infected. When $\gamma < \beta$ the number of infected stabilises at some finite fraction of the population and the disease is said to be **endemic**. When $\gamma > \beta$ recovery happens faster than infection and the disease dies out completely.

7.3 SIR model

Now we add recovered and immune (or dead) individuals in state R . We have three coupled equations

$$\begin{aligned}\frac{ds}{dt} &= -\beta si \\ \frac{di}{dt} &= \beta si - \delta i \\ \frac{dr}{dt} &= \delta i\end{aligned}\tag{7}$$

with $s + i + r = 1$.

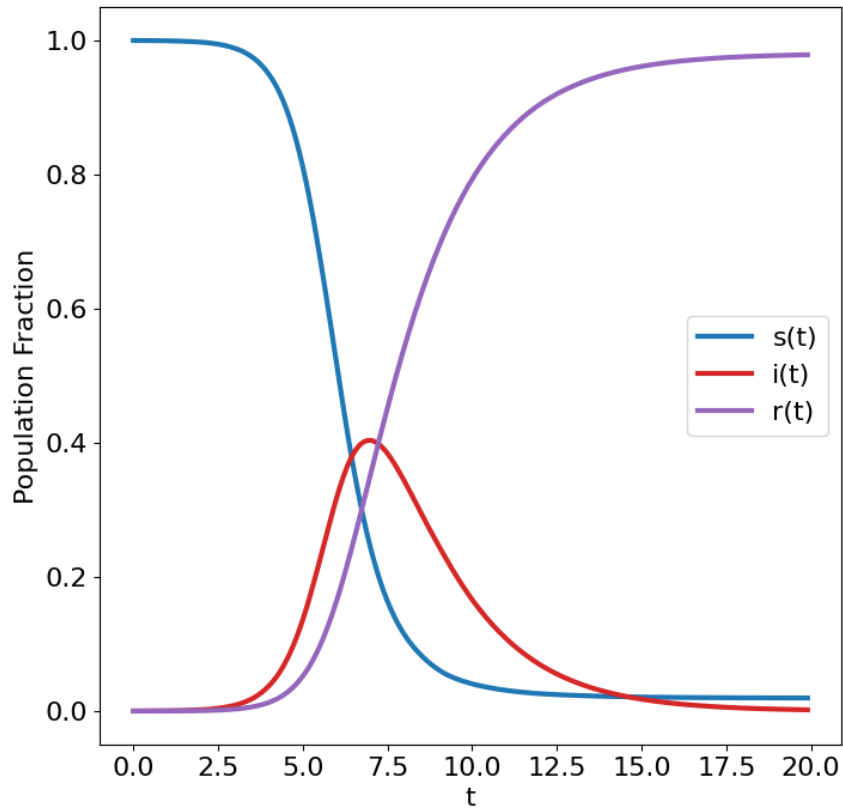


Figure 3: Solution of the SI model with $\beta = 2$, $\delta = 0.5$.

This can't be solved analytically (the integral is too hard) but the behaviour is illustrated in Figure 3. As you might expect, in this model everyone eventually ends up in the R state

7.4 SIRS model

For completeness, the SIRS model allows recovered individuals can become susceptible again at some rate δ . Thus

$$\begin{aligned}\frac{ds}{dt} &= -\beta si + \gamma r \\ \frac{di}{dt} &= \beta si - \delta i \\ \frac{dr}{dt} &= \delta i - \gamma r\end{aligned}\tag{8}$$

This is similar to the SIS model, but with an extra step.

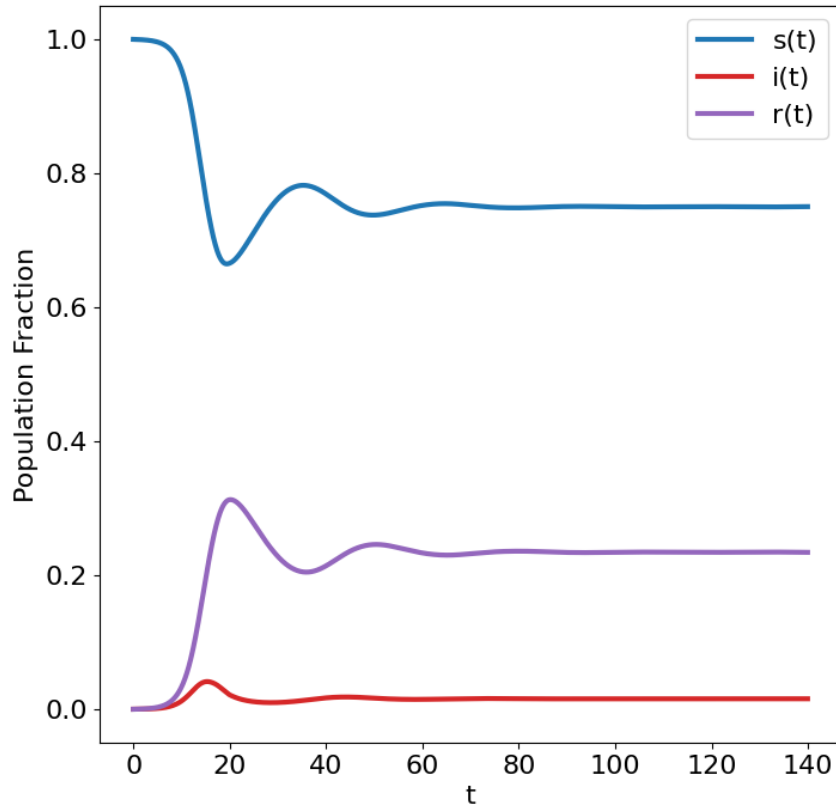


Figure 4: Solution of the SI model with $\beta = 2$, $\delta = 0.5$, $\gamma = 0.1$.

Figure 4 an example for one choice of parameters where the infection vanishes, $i \rightarrow 0$.

7.5 Reproductive number

In the SI model, the **characteristic time** to infect a fraction $1/e \sim 37\%$ of all the individuals is

$$\tau = \frac{1}{\beta} \quad (9)$$

A similar characteristic time can be computed for all models. The SIS model is interesting, in that there are two possible end states, one where the disease

is present in some finite fraction of the population (endemic) and one where it is absent. Setting $\frac{di}{dt} = 0$ for the SIS equations gives

$$i = 1 - \frac{\gamma}{\beta}$$

note if $\gamma > \beta$ then this is *negative*. Since we can't have a negative population this implies that in this case, with a sufficiently high recovery rate, the disease disappears. The ratio

$$R_0 = \frac{\beta}{\gamma}$$

is called the **basic reproductive number**. It is the average number of susceptible individuals infected by an I individual during their infectious period in a fully susceptible population. If $R_0 > 1$ then the disease is endemic in the long run, if it is less than 1 then the disease eventually vanishes.

7.6 SI model on a network

We will study the simplest model, SI, on a network. The basic intuition is that individuals with more contacts (higher degree) should be more easily infected, as well as being more likely to infect others. We replace the β in the homogeneous models above with βk , the rate of infection times the degree. We can make some progress with analytical investigation of the model using a trick called the **degree block approximation**. Set

$$i = \sum_k p_k i_k$$

where i_k is the probability for infection among all degree k nodes.

$$i_k = \frac{I_k}{N_k}$$

We then write down the SI equation for each degree

$$\frac{di_k}{dt} = \beta k(1 - i_k)\theta_k \quad (10)$$

note we replaced β with βk , and the second i with θ_k , which is the fraction of infected neighbours of a degree k node. The θ_k term determines how the degrees interact with each other.

We seem to have made our life much more difficult, going from one equation to many coupled ones. We make progress with the very simplifying assumptions that there are **no degree correlations**. This means θ_k is independent of k . We also assume that we are only interested in the characteristic time, so we only need to study the epidemic in its early phase when i_k is small. Since both i_k and θ_k are small in this regime $i_k\theta_k$ is negligible and we drop it to get

$$\frac{di_k}{dt} = \beta k \theta \quad (11)$$

where we have also dropped the dependence of θ on k .

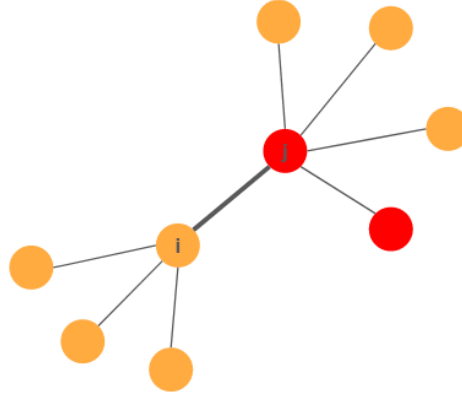


Figure 5: A random infected neighbour of i has k additional edges, one of which is infected.

Now we want an expression relating θ and i . In the SI model we assume that an infected node is connected to just one other infected node, the one that infected it, see Figure 5, remember we are at early times when most of the network is still S. In the absence of degree correlations the probability a random neighbour has remaining degree k node is proportional to

$$(k+1)p_{k+1}$$

normalizing, this gives

$$q_k = \frac{(k+1)p_{k+1}}{\langle k \rangle}$$

If the node is infected then we have to use up one of its links to connect to another infected so

$$q'_k = \frac{kp_{k+1}}{\langle k \rangle}$$

The proportion of those which are infected is $q'_k i_{k+1}$ and we can get the total number of infected by summing over all degrees

$$\theta = \sum_k \frac{k p_{k+1}}{\langle k \rangle} i_{k+1} = \sum_k \frac{(k-1) p_k}{\langle k \rangle} i_k$$

This is independent of k as required. Differentiating gives

$$\frac{d\theta}{dt} = \sum_k \frac{(k-1) p_k}{\langle k \rangle} \frac{di_k}{dt} \quad (12)$$

Subbing in to the simplified SI model

$$\begin{aligned} \frac{d\theta}{dt} &= \sum_k \frac{(k-1) p_k}{\langle k \rangle} (\beta k \theta) \\ \frac{d\theta}{dt} &= \beta \left(\frac{\langle k^2 \rangle}{\langle k \rangle} - 1 \right) \theta \end{aligned} \quad (13)$$

which has the solution

$$\theta = C e^{t/\tau} \quad (14)$$

where the characteristic time τ is

$$\tau^{SI} = \frac{\langle k \rangle}{\beta(\langle k^2 \rangle - \langle k \rangle)} \quad (15)$$

For a *random network* $\langle k^2 \rangle = \langle k \rangle(\langle k \rangle + 1)$ giving

$$\tau_{ER}^{SI} = \frac{1}{\beta \langle k \rangle}$$

the same result as we found for the homogeneous network.

More interesting, for a network with a long tail, in particular a power law with exponent less than 3, in the infinite N limit $\langle k^2 \rangle \rightarrow \infty$ (think about the integral $\int_0^\infty k^2 k^{-2} dk$). This infinity in the denominator makes $\tau \sim 0$ i.e. infection spreads instantaneously!

7.7 SIS model on a network

The same substitutions for the SIS model give

$$\tau^{SIS} = \frac{\langle k \rangle}{\beta \langle k^2 \rangle - \gamma \langle k \rangle} \quad (16)$$

To have an epidemic we need $\tau > 0$. This condition is usually expressed in terms of the **spreading rate**

$$\lambda = \frac{\beta}{\gamma} \quad (17)$$

This will be high for diseases which are easily contracted, high β , and have long recovery times, low γ . The **critical spreading rate**, λ_c is the value of λ at which τ changes sign.

$$\tau^{SIS} > 0 \implies \frac{\beta}{\gamma} = \lambda > \frac{\langle k \rangle}{\langle k^2 \rangle} \quad (18)$$

So the critical spreading rate is

$$\lambda_c = \frac{\langle k \rangle}{\langle k^2 \rangle} \quad (19)$$

For a random network again $\langle k^2 \rangle = \langle k \rangle (\langle k \rangle + 1)$ and

$$\lambda_c^{ER} = \frac{1}{\langle k \rangle - 1} \quad (20)$$

However again, for sufficiently fat tailed networks $\langle k^2 \rangle$ diverges and $\lambda_c \simeq 0$ so even diseases which are difficult to spread and easy to recover from can become endemic.

7.8 Immunization

If we have some vaccine or cure for the disease we can use it to make some nodes invisible to the epidemic process. If people are immunized at random let g be the proportion of the population who are immunized. We can incorporate this in to β as $\beta(1 - g)$, which measures randomly chosen contacts with non-immunized individuals. Now, for random immunization on a random network we have the critical spreading rate

$$(1 - g_c)\lambda = \frac{1}{\langle k \rangle - 1}$$

Solving for g_c

$$g_c = 1 - \frac{\gamma}{\beta} \frac{1}{\langle k \rangle - 1} \quad (21)$$

If the vaccination proportion goes above this value, the spreading goes below the epidemic threshold and it dies out. Properties of the disease which make this easier are high recovery γ , low spreading β and properties of the network

which make vaccination easier are small $\langle k \rangle$ i.e. few connections. All as you would expect. In general

$$g_c = 1 - \frac{\gamma \langle k \rangle}{\beta \langle k^2 \rangle} \quad (22)$$

Again, for networks where $\langle k^2 \rangle$ diverges we have $g_c \simeq 1$, meaning we have to immunize basically the entire population to eliminate the disease.

In such situations we could decrease the $\langle k^2 \rangle$ factor substantially by targeting the high-degree nodes. A clever strategy to do this relies on the **friendship paradox**, the fact that on average the neighbours of a node have higher degree than the node itself.

- Randomly select a group of nodes, call these Group 0
- Select a random friend of each member of Group 0 and call this Group 1 (obviously friends in common only go in once!)
- Immunize the Group 1 individuals

This requires no information about the topology of the network but disproportionately targets high degree nodes!

References

- [1] Márton Pósfai and Albert-László Barabási. *Network science*. Cambridge University Press, 2016.
- [2] Mark Newman. *Networks*. Oxford university press, 2018.